

Lawrence R. Liu

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EDUCATION

University of California Los Angeles

Masters of Science— Electrical Engineering— GPA: 4.0/4.0

Los Angeles, CA

Sep 2024 – June 2026

University of California Los Angeles

BS, Electrical Engineering, GPA: 3.8/4.0

Los Angeles, CA

Sep. 2021 – June 2024

RESEARCH EXPERIENCE

Graduate/Undergraduate Researcher

April 2023 – Present

Advised By Professor Lin F. Yang

UCLA

- Conducting research on efficient LLM inference through compression and length aware reasoning.
- Developed the **NoWag** family of shape preserving compression algorithms, Accepted as a first author paper at the *Conference on Language Modeling (COLM)*
- Showed that the 2 bit Vector Quantization NoWag (**NoWag-VQ**) algorithm, outperforms SOTA methods while requiring 48x less calibration data.
- Developed **ARMOR**, a novel one-shot semi-structured pruning algorithm that factorizes LLM weight matrices into a hardware-efficient 2:4 sparse core with adaptive block-diagonal wrappers
- Achieved state-of-the-art performance–efficiency trade-offs on Llama and Qwen models, reducing performance loss by up to 50% while maintaining inference speedups and memory savings

Undergraduate Researcher

February 2022 – October 2024

Advised By Professor Vwani Roychowdhury

UCLA

- Developed data driven Deep Learning algorithms to guide treatment for epilepsy patients
- Trained and pruned a lightweight deep learning classifier for epilepsy biomarker detection
- Built unsupervised time-frequency VAEs and graph-based models in PyTorch to discover more accurate, faster to detect, biomarkers for predicting surgery outcomes
- Led a multidisciplinary team to build a Python GUI tool for real-time brain signal recording and biomarker detection, now actively used by pediatric neurologists at UCLA
- Results published at *Journal of Neural Engineering* and *Epilepsia*

TEACHING EXPERIENCE

Teaching Assistant

April 2024 – June 2024

UCLA Electrical and Computer Engineering Department

Los Angeles, CA

- TAed for a graduate class, Deep Learning 2 (ECE 239AS) as an undergraduate for spring quarter 2024, lead office hours and discussions
- Created from scratch two Deep Reinforcement Learning projects in python with Deep Deterministic Policy Gradient (DDPG) and Deep Q Learning (DQN)

PROJECTS

Kernel-Density-Estimation Stock Trading Algorithm

April 2023 - June 2023

- Developed a novel Kernel-Density based algorithm to identify optimal portfolio weights for a buy and hold strategy in stocks
- Lead a team of 4 representing UCLA against 40+ teams from Stanford, Harvard, CMU, etc at the UChicago 2023 Trading Competition
- Optimized Kernel Parameters with Stochastic Gradient Descent for greater accuracy, efficiency, and scalability
- Utilized Spectral Clustering to identify sectors of correlated stocks, iterative applied this algorithm on each sector
- Backtested on 2 years of real world S&P 500 data and was able to achieve a 17.4% year over year return and a Sharpe ratio of 1.424, outperformed the S&P500 and naive Markowitz Baseline

PUBLICATIONS

Journal/Conference Papers

- **Lawrence Liu**, Inesh Chakrabarti, Yixiao Li, Mengdi Wang, Tuo Zhao, Lin F. Yang. “NoWag: A Unified Framework for Shape Preserving Compression of Large Language Models”, *Conference of Language Modeling (COLM)*, 2025, [Accepted]
- Yipeng Zhang, Atsuro Daida, **Lawrence Liu**, et al. “Self-Supervised Data-Driven Approach Defines Pathological High-Frequency Oscillations in Epilepsy”, *Epilepsia*, 2025 [Accepted]
- Yipeng Zhang, **Lawrence Liu**, Yuanyi Ding, et al. “PyHFO: Lightweight Deep Learning-powered End-to-End High-Frequency Oscillations Analysis Application”, In *Journal of Neural Engineering* , 21, 2024.

Preprints

- **Lawrence Liu**, Alexander Liu, Mengdi Wang, Tuo Zhao, Lin F. Yang, “ARMOR: High-Performance Semi-Structured Pruning via Adaptive Matrix Factorization”, 2025.
- Chang Liu, Yiran Zhao, Yaoqi Ye, **Lawrence Liu**, Csaba Szepesvari, Lin F. Yang, “LACONIC: Length-Aware Constrained Reinforcement Learning for LLM”, 2025
- Tonmoy Monsoor, Atsuro Daida, Prateik Sinha, Yipeng Zhang, **Lawrence Liu**, et al. , “Synchronization network-based approach for accurate epileptogenic zone identification from short interictal EEG data ”, 2025.

Workshop Papers

- Preliminary Results of the NoWag framework presented at **ICLR 2025 Sparsity in LLMs Workshop**

AWARDS & HONORS

Dean’s List — *UCLA Henry Samueli School of Engineering* — Fall 2021, Fall 2022, Winter 2023, Spring 2023, Spring 2024

Eta Kappa Nu (HKN) Honor Society — *UCLA* — Fall 2023

TECHNICAL SKILLS

Languages: Python, C/C++, R, Matlab, Octave, Julia, Mathematica

Frameworks & Tools: Git, Bash Scripting, Pytorch, Sklearn, Tensorflow, Pandas, NumPy, Matplotlib, Simulink, Gymnasium, Optuna, Cvxpy, Mosek, Cuda, Transformers

Specialized Techniques:

- **Machine Learning:** Deep Learning, Reinforcement Learning, Graph Neural Networks, Generative AI, Model Parallelism
- **Signal Processing:** Time Series Analysis, Statistical Analysis, Data Mining, Data Visualization
- **Optimization:** Numerical Analysis, Convex Optimization
- **Systems:** Control Theory, High-Performance Computing